

The Complete Love-Mode Threshold Ladder for Kirchhoff's Uniform Elliptic Vortex

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Abstract

For a uniform-vorticity ellipse in the unbounded plane, we separate the classical input from a complete all-mode consequence. The patch is an exact Euler relative equilibrium with shape rate $\Omega = \omega_0 ab / (a + b)^2$. Love's dispersion relation factors into two dimensionless scalar terms: one is positive and the other has exactly one zero for every mode $m \geq 3$. We prove strict ordering of all thresholds, the sharp first wall $a/b = 3$, and an exact closed asymptotic description of the ladder. Exact rational evidence audits 35,904 modal cells. Every stability statement is spectral and linear; no nonlinear stability, Jordan, or filamentation theorem is asserted.

1 The exact Euler relative equilibrium

Consider the planar Euler vorticity equation

$$\partial_t \omega + u \cdot \nabla \omega = 0, \quad \nabla \cdot u = 0, \quad \text{curl } u = \omega, \quad (1)$$

with velocity decaying modulo its circulation field at infinity. Let

$$E_0 = \left\{ (x, y) : \frac{x^2}{a^2} + \frac{y^2}{b^2} < 1 \right\}, \quad a \geq b > 0, \quad \omega(x, 0) = \omega_0 \mathbf{1}_{E_0}. \quad (2)$$

Positive ω_0 is counterclockwise.

Theorem 1 (Kirchhoff shape clock). *For $a > b$, the Kirchhoff patch evolution from (2) is the relative equilibrium*

$$E_t = R_{\Omega t} E_0, \quad \Omega = \frac{\omega_0 ab}{(a + b)^2}. \quad (3)$$

The interior velocity at the displayed orientation is

$$u(x, y) = \frac{\omega_0}{a + b} (-ay, bx). \quad (4)$$

The area, circulation, and quadratic vorticity moment are

$$|E_t| = \pi ab, \quad \Gamma = \pi \omega_0 ab, \quad I := \int_{\mathbb{R}^2} |x|^2 \omega(x, t) dx = \frac{\pi \omega_0 ab (a^2 + b^2)}{4}. \quad (5)$$

If $\Omega \neq 0$, the unmarked patch has minimal period $\pi/|\Omega|$; an oriented-axis lift has period $2\pi/|\Omega|$.

Proof. With $u = (-\psi_y, \psi_x)$, the interior streamfunction is

$$\psi_{\text{in}} = \frac{\omega_0}{2(a + b)} (bx^2 + ay^2),$$

which gives (4), zero divergence, and $\text{curl } \omega_0$. In elliptic coordinates $x = c \cosh \xi \cos \eta$, $y = c \sinh \xi \sin \eta$, the harmonic exterior expression

$$\psi_{\text{out}} = \frac{\omega_0 ab}{2} \xi + \frac{\omega_0 ab}{4} e^{-2\xi} \cos 2\eta + C$$

matches the boundary Cauchy data for a suitable constant C ; this is the classical Kirchhoff construction [2]. At $(a \cos \theta, b \sin \theta)$, dotting the difference between (4) and $\Omega(-y, x)$ with the unnormalized normal $(\cos \theta/a, \sin \theta/b)$ gives

$$\sin \theta \cos \theta \left\{ \frac{\omega_0(a-b)}{a+b} - \frac{\Omega(a^2-b^2)}{ab} \right\}.$$

It vanishes exactly for (3); tangential differences only reparametrize the contour. The contour therefore rotates without changing shape. Direct ellipse integration gives (5). Finally, $R_\pi E_0 = E_0$ while no smaller positive rotation fixes a noncircle, proving the two period statements. $\square$

References

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